Normal PEPS Fundamental Theorem: Section 3 Route

2026-05-06

1 The source argument

The normal-PEPS part of [MGRPG$^+$18] begins after the injective PEPS theorem. Its purpose is not to replace the injective proof, but to make the injective proof applicable after a controlled blocking. A normal PEPS is one for which sufficiently large regions can be blocked into injective tensors. Once such injective regions are available, the proof again applies the three-site injective-chain argument around a chosen edge.

The source proof has three layers. First, Lemma `injective_union` proves that the union of two injective regions is injective. The proof blocks the network into the four regions $A \setminus B$, $A \cap B$, $B \setminus A$, and $(A \cup B)^c$. If a boundary tensor kills the contraction over $A \cup B$, injectivity of A first removes the $A \setminus B$ and $A \cap B$ part, and injectivity of B then removes the remaining $A \cap B$ and $B \setminus A$ part. [1]

Second, for the translationally invariant square-lattice theorem, the paper assumes that every 2×3 and 3×2 region is injective. By repeatedly applying the union lemma, the regions denoted R, S, and T in the paper are injective. The finite-lattice block denoted A_3 in the proof of Theorem 3 has the same complementary shape as T but lives in the full finite lattice around the chosen edge; the present coordinate definition records only the displayed local-window complement. The regions R and S differ by one site, and their complements are injective when the system is large enough. These are the regions used in the last comparison step. [2]

Third, the proof of Theorem 3 blocks a neighborhood of each edge into three injective pieces. The red region is A_1, the blue region is A_2, and the complement is A_3. This produces the same three-site injective MPS configuration used in the injective PEPS proof. Lemma `inj_isomorph` therefore gives a gauge on every edge. Translation invariance reduces these edge gauges to one horizontal

[1] The source is `Papers/1804.04964/paper_normal.tex:1324--1400`.

[2] The regions R, S, and T are drawn at `Papers/1804.04964/paper_normal.tex:1407--1443`.

matrix X and one vertical matrix Y. After absorbing them into the second tensor, the final comparison of the R and S regions gives $A \propto \widetilde{B}$. [3]

The general theorem `normal` removes the square-lattice specialization. It assumes precisely the blockings needed above: every edge can be blocked into a three-site injective chain, and every site admits two injective regions with injective complements that differ exactly in that site. [4]

2 Current formal representation

The injective PEPS objects used by the normal theorem are represented in `TNLean/PEPS/Defs.lean`, `TNLean/PEPS/VirtualInsertion.lean`, and `TNLean/PEPS/Blocking.lean`. Chapter 13a of the blueprint has a separate normal-PEPS section after the injective theorem. This separation is important: the normal theorem needs extra blocking hypotheses before the injective proof route can be invoked.

The present blueprint nodes are the following.

- `lem:peps_injectiveRegion_union` states the union-of-injective regions lemma. It is tracked by GitHub issue #1374. It records the four-region decomposition and the two successive inverse applications used in the source proof.
- `lem:peps_regionInjectivityUnionClosure_finite` records the finite nonempty iteration of the binary union-closure assertion.
- `def:peps_normalSquareBlockingRegions` records the square-lattice regions R, S, and T. It is part of GitHub issue #1375. The coordinate rectangular hypotheses, union closure, and a supplied rectangular cover of the displayed T-region imply this abstract structure.
- `thm:peps_normalEdgeBlockingToInjectiveChain` states that the normal square-lattice hypotheses allow a blocking around every edge into three injective regions. It is part of GitHub issue #1375.
- `thm:peps_tiNormalGaugeAbsorption` records the translationally invariant gauge absorption with horizontal gauge X and vertical gauge Y. It is part of GitHub issue #1375.
- `thm:fundamentalTheorem_normalSquarePEPS` records the square-lattice theorem corresponding to Theorem 3. It is part of GitHub issue #1375.
- `thm:fundamentalTheorem_normalPEPS` records the general normal PEPS theorem following Theorem 3. It is part of GitHub issue #1375.

[3] The proof of Theorem 3 is `Papers/1804.04964/paper_normal.tex:1449--1572`.
[4] The general theorem is `Papers/1804.04964/paper_normal.tex:1574--1585`.

The finite rectangular square lattice is represented by pairs of horizontal and vertical coordinates. Its graph has horizontal and vertical nearest-neighbor edges, named coordinate right/up edges, and orientation predicates distinguishing the two possible directions. Every edge has exactly one of these orientations. With the ordered endpoints convention for edges, horizontal edges are coordinate right edges from their left endpoints, and vertical edges are coordinate upward edges from their lower endpoints.

The four-region decomposition from Lemma `injective_union` is formalized both by named regions and by an indexed family in the source-paper order: $A \setminus B$, $A \cap B$, $B \setminus A$, and $(A \cup B)^c$. The indexed family reconstructs A, B, $A \cup B$, A^c, B^c, and the full vertex set. The same reconstructions are available as finite unions over the selected indexed subfamilies $\{0, 1\}$, $\{1, 2\}$, $\{0, 1, 2\}$, $\{2, 3\}$, and $\{0, 3\}$. The singleton subfamily $\{3\}$ gives the outside block and is the complement of the selected inside subfamily $\{0, 1, 2\}$. Its selected complementary pairs match the two places where the source proof applies injectivity of A and then injectivity of B. The selected subfamilies $\{0, 1\}$ and $\{2, 3\}$ are disjoint, as are $\{1, 2\}$ and $\{0, 3\}$; these are the disjointness forms needed for the two inverse applications. The abstract union-closure hypothesis also has the corresponding indexed inside-union form: injectivity of the selected subfamilies $\{0, 1\}$ and $\{1, 2\}$ implies injectivity of the inside subfamily $\{0, 1, 2\}$. Every vertex belongs to exactly one indexed piece. These facts prepare the tensor-level boundary-contraction proof, but do not prove that theorem.

The rectangular vocabulary separates coordinate products from source-paper contiguous blocks. Coordinate rectangles are products of finite coordinate sets, whereas contiguous coordinate rectangles are products of coordinate intervals. The 2×3 and 3×2 coordinate-product predicates assert only the two cardinalities; the contiguous-rectangle predicates express the source-paper blocks. The square-lattice rectangular injectivity hypotheses use these two contiguous predicates and map to the abstract rectangular-injectivity bundle. Bounded contiguous rectangles of sizes 2×3 and 3×2 satisfy the corresponding source-rectangle predicates.

The displayed regions R and S are explicit unions of contiguous rectangles. Region R is the union of one contiguous 2×3 rectangle and one contiguous 3×2 rectangle. Region S is the union of two contiguous 2×3 rectangles. Their coordinate definitions agree with the source claim that they differ in one site: $R \subseteq S$, and $S \setminus R$ is the lower-right site of the displayed 3×3 square. Assuming the union-closure assertion supplied by the source lemma on unions of injective regions, these decompositions imply injectivity of R and S. The formal development also contains the finite nonempty iteration of this union-closure assertion, matching the repeated unions used after Lemma `injective_union`.

The displayed local model for T is the complement, inside the source 5×6 window, of the two edge blocks in the source picture. These removed blocks are one contiguous 2×3 rectangle and one contiguous 3×2 rectangle, hence they are injective under the coordinate rectangular hypotheses. They are disjoint, and together with the local-window T region they reconstruct the 5×6 window. This local-window region is distinct from the finite-lattice complementary block A_3

used later in the proof of Theorem 3. The finite-lattice block is the complement of the two edge blocks in the full square lattice; its elementary disjointness and covering identities are formalized separately.

A target-parametric rectangular-cover criterion supplies a sufficient injectivity route: if a square-lattice region has a nonempty cover by contiguous 2×3 and 3×2 rectangles, then rectangular injectivity and finite union closure imply injectivity of that region. The formal special cases are the local-window T region and the finite-lattice complementary block. This criterion is only a sufficient local model, not the source argument itself.

The origin-based local-window model for T has no such rectangular cover whenever the ambient lattice contains the corresponding 5×6 window. The obstruction is a point-forcing argument. For the local-window target T_{loc} at the origin, let $R^{a\times b}_{x,y}$ denote the contiguous $a \times b$ rectangle with lower-left corner (x, y). Then

$$p := (0,0) \in T_{\mathrm{loc}}, \tag{1}$$

$$p \in R^{2\times 3}_{x,y} \implies q_{23} := (0,2) \in R^{2\times 3}_{x,y}, \qquad q_{23} \notin T_{\mathrm{loc}}, \tag{2}$$

$$p \in R^{3\times 2}_{x,y} \implies q_{32} := (2,1) \in R^{3\times 2}_{x,y}, \qquad q_{32} \notin T_{\mathrm{loc}}. \tag{3}$$

Thus any exact cover of T_{loc} by 2×3 and 3×2 rectangles would cover p by one of its pieces and would therefore also cover a point outside T_{loc}. The same calculation applies to the rotated local-window model for vertical edges in a 6×5 window, to the origin-based horizontal edge-complement model in the 5×7 frame, and to the origin-based vertical edge-complement model in the 7×5 frame. The forced points are

target	p	q_{23}	q_{32}
T_{loc}	$(0,0)$	$(0,2)$	$(2,1)$
$T^{\mathrm{rot}}_{\mathrm{loc}}$	$(0,0)$	$(1,2)$	$(2,1)$
$A_{3,\mathrm{hor}}$	$(0,0)$	$(0,2)$	$(2,1)$
$A_{3,\mathrm{ver}}$	$(0,0)$	$(1,2)$	$(2,1)$

The formal point-forcing lemma is TNLean.PEPS.not_squareLatticeRectangleCover_of_forced_points. Its four present applications are the corresponding no-cover theorems for the local T models and the edge-complement models. These no-cover results do not contradict the paper. They show that the source's injectivity argument for T and A_3 must be aligned with the finite PEPS geometry, rather than read directly from the present exact-cover criterion.

The normalized horizontal edge frame has a collar identity: in the 5×7 frame, A_3 is the local-window T region together with two top-collar contiguous 3×2 rectangles. Thus local T-injectivity implies injectivity of A_3 by union closure and rectangular injectivity. The vertical counterpart is a 7×5 frame in which the edge complement is the rotated local T region together with two right-collar contiguous 2×3 rectangles. A rectangular cover of the rotated local T region therefore implies injectivity of the vertical complementary block.

The normalized horizontal and vertical edge blockings give one-edge data with red, blue, and complementary regions. The distinguished edges have the expected orientations; their endpoints lie in the red and blue regions; the three regions are pairwise disjoint and cover the finite lattice. This one-edge datum is the local form of the later every-edge blocking hypotheses. A family of such one-edge data determines those hypotheses, and each datum supplies the three injective regions used in the associated edge-blocked chain.

The translated horizontal and vertical pictures carry the same data at an arbitrary coordinate origin. When the translated frame fits in the finite lattice, the translated edge is identified with the corresponding coordinate right or upward edge. A rectangular cover of the translated complementary region supplies the one-edge blocking datum and hence the three injective regions. The normalized pictures are the origin specializations of these translated pictures, while the normalized and translated data remain in separate Lean modules.

The conditional every-edge construction has two equivalent forms. A translated edge-window assignment gives, for every edge, a horizontal or vertical window and a rectangular cover of its complement. Such an assignment implies the normal edge-blocking hypotheses. An oriented margin-and-cover datum records the same input directly on each square-lattice edge: the edge orientation, the coordinate inequalities placing the translated frame, and the complementary cover. A family of these data also determines the normal edge-blocking hypotheses, and the resulting hypotheses recover the supplied one-edge datum at each edge.

For the present open rectangular coordinate graph, the margin predicates describe only interior translated frames. A coordinate right edge must have enough left and right margin; a coordinate upward edge must have enough bottom and top margin. Hence neither the translated-window criterion nor the oriented margin-and-cover criterion supplies data on all edges of the open 7×7 graph. Explicit boundary obstructions are recorded on all four sides. The remaining every-edge construction must therefore include the boundary geometry and the source-faithful local and rotated T covers.

Verdict: this note records an open gap with a scope restriction. The earlier vocabulary gap for contiguous rectangular blocks is closed. The earlier coordinate rectangular-injectivity structure is also present, and the indexed finite-region geometry of Lemma `injective_union` is formalized down to the selected complement and closure forms used by the two inverse applications. The rectangular decompositions of R and S are formalized, and the conditional derivations of injectivity for R and S from union closure are also formalized. The two edge blocks removed from the displayed T-window are injective under the coordinate rectangular hypotheses. The finite-lattice complementary block around the chosen edge has a set-theoretic definition. The exact-cover criterion by 2×3 and 3×2 rectangles is only a restricted sufficient condition, and the origin-based instances above show that it is not the source's injectivity argument. The source-faithful covers and the every-edge blocking construction are not yet derived. The remaining open gap is to prove the tensor-level union theorem, realign the local T-injectivity argument with the finite PEPS geometry of the source, translate the collar decomposition around each edge, and then prove the unconditional injectivity of R, S, and the edge-complementary block.

The normal route itself is tracked by GitHub issue #1373. The normal-section diagram work is tracked by GitHub issue #1376. The general PEPS status tracker is GitHub issue #1252.

3 Remaining mathematical obligations

The normal PEPS theorem cannot be obtained by citing the injective theorem alone. The lists below are the authoritative record for the square-lattice normal blocking route. Lean docstrings and blueprint entries should point here rather than repeat this status in full.

The following part of the formalization is already present.

- Region injectivity is represented by TNLean.PEPS.RegionInjectivityData; the same infrastructure contains the region-complement definitions TNLean.PEPS.regionComplement and TNLean.PEPS.regionOutsideUnion, and the finite-region union-closure hypothesis TNLean.PEPS.RegionInjectivityUnionClosure.
- The four-region decomposition for Lemma `injective_union` is formalized both by named regions and by the indexed family $A \setminus B$, $A \cap B$, $B \setminus A$, and $(A \cup B)^c$, including the reconstruction, disjointness, and selected-complement identities used by the two inverse applications.
- The abstract contraction-chain criterion for Lemma `injective_union` is formalized: if the three implications

  $$\mathcal{C}_{\{0,1,2\}}(X) = 0 \Longrightarrow \mathcal{C}_{\{2\}}(X) = 0 \Longrightarrow \mathcal{C}_{\{1,2\}}(X) = 0 \Longrightarrow X = 0$$

  are supplied, then the union contraction has trivial kernel.
- The square-lattice coordinate model, contiguous 2×3 and 3×2 rectangles, and the coordinate rectangular-injectivity hypotheses are present.
- The source regions R and S are explicit finite coordinate regions, their rectangular decompositions are formalized, and their injectivity follows conditionally from rectangular injectivity and union closure.
- The displayed local T-region, the finite-lattice edge-complement regions, their sufficient rectangular-cover criteria, the no-cover diagnostics for the present origin-based local models, and the normalized collar decompositions are formalized.
- The one-edge, translated-window, and margin-cover hypotheses for normal edge blocking are present, together with their endpoint, disjointness, coverage, and injective-chain consequences.

The remaining obligations are the following.

1. Prove the tensor-level union-of-injective-regions lemma. The proof must follow the source proof by applying the left inverses for A and B in sequence.

2. Realign the source proof of injectivity for the local and rotated T-regions with the finite PEPS geometry. The present exact-cover criterion is a useful sufficient condition, but the recorded point-forcing obstructions show that it is not the source argument for the origin-based local models.

3. Prove the every-edge finite-geometry construction. The normalized and translated one-edge data give the desired red, blue, and complementary injective regions when the corresponding window or margin-cover datum is supplied; the remaining task is to construct source-faithful data for every edge, including boundary edges.

4. Reuse the injective-chain insertion route after this blocking: GitHub issue #1367, GitHub issue #1368, GitHub issue #1363, GitHub issue #1364, GitHub issue #1361, and GitHub issue #1360.

5. In the translationally invariant case, show that the edge gauges obtained after blocking reduce to one horizontal gauge X and one vertical gauge Y, up to scalar.

6. Prove the scalar condition $\lambda^{nm} = 1$ for the square-lattice theorem, and state separately the scalar freedom in the non-translation invariant normal theorem.

4 Source-to-formalization ledger

- Lemma `injective_union`: blueprint `lem:peps_injectiveRegion_union`, issue #1374.
- Finite iteration of Lemma `injective_union`: blueprint `lem:peps_regionInjectivityUnionClosure_finite`, issue #1374.
- Regions R, S, and T: blueprint `def:peps_normalSquareBlockingRegions`, issue #1375. This node also records the conditional implication from coordinate rectangular injectivity, union closure, and a rectangular cover of the displayed T-region to the abstract R, S, T blocking-region structure.[5]
- Injectivity of the two edge blocks removed from the displayed T-window, and of their union under union closure: blueprint `lem:peps_normalSquareRegionT_edgeBlocks_injective`, issue #2004.
- General rectangular-cover criterion for square-lattice regions: blueprint `lem:peps_normalSquareRectangleCover_injective`, issue #2004.
- Rectangular-cover criterion for injectivity of the displayed T-region: blueprint `lem:peps_normalSquareRegionT_injective_of_cover`, issue #2004. This node also contains the rotated vertical local T-region cover criterion.

[5] The formal declaration is TNLean.PEPS.normalSquareBlockingRegions˙of˙TCover.

- Obstruction to applying that criterion directly to the present origin-based local-window T models and edge-complement models: blueprint `lem:peps_normalSquareRegionT_no_five_by_six_cover`, issue #2004.

- Finite-lattice complementary block around an edge: blueprint `lem:peps_normalSquareEdgeComplementRegion`, issue #2004.

- Rectangular-cover criterion for injectivity of that block: blueprint `lem:peps_normalSquareEdgeComplementRegion_injective_of_cover`, using the cover definition `def:peps_normalSquareRectangleCover`, issue #2004.

- Extension of a local T rectangular cover to a rectangular cover of the normalized horizontal-edge complementary block by adjoining the two top-collar rectangles: blueprint `lem:peps_normalSquareEdgeComplementRegion_coverOfT`, issue #2004.

- Normalized 5×7 top-collar decomposition of that block: blueprint `lem:peps_normalSquareEdgeComplementRegion_topCollar`, issue #2004. This node also contains a normalized 7×5 collar identity for the analogous rectangular frame and the rotated vertical edge-complement collar identity.

- Normalized horizontal and vertical red/blue/complement blocking data: blueprint `lem:peps_normalSquareHorizontalEdge_blockingData`, issue #2004. These use the common one-edge blocking-data structure. They also expose the corresponding injective-chain conclusions. The horizontal complement injectivity is reduced to a local T-cover, and the vertical complement injectivity is reduced to a rotated local T-cover.

- Translated horizontal red/blue/complement blocking data: blueprint `lem:peps_normalSquareHorizontalTranslatedEdge_blockingData`, issue #2004. This is the coordinate-origin-parametric horizontal step toward the every-edge construction. Its coordinate origin specializes to the normalized horizontal picture.

- Translated vertical red/blue/complement blocking data: blueprint `lem:peps_normalSquareVerticalTranslatedEdge_blockingData`, issue #2004. This is the coordinate-origin-parametric vertical step toward the every-edge construction. Its coordinate origin specializes to the normalized vertical picture.

- Conditional construction from translated edge windows: blueprint `thm:peps_normalSquareTranslatedEdgeBlockingHypotheses_of_windows`, issue #2004. This records the formal boundary before proving that every edge in the finite square lattice admits such a translated window. Coordinate right and upward edges are tied to the corresponding translated windows, and the margin hypotheses for actual coordinate right/up edges are explicit. Arbitrary horizontal and vertical edges inherit these criteria from their coordinate representatives. The oriented margin-and-cover data structure records the remaining per-edge input in the open rectangular coordinate model. One such datum directly supplies one-edge blocking data and the corresponding injective-chain conclusion, and a family of such data determines the normal edge-blocking hypotheses. The resulting hypotheses recover the supplied datum and its three injective regions at each edge. The horizontal and vertical margin predicates are named, the coordinate right/up and oriented-edge window constructors accept those predicates directly, translated windows on coordinate right/up edges force those predicates, and the open 7×7 graph does not admit a family of these translated windows over all edges; four explicit boundary examples are recorded as a single conjunction. The per-edge margin-cover structure stores the same predicates, and the resulting open-boundary obstruction for horizontal edges without enough left or right margin and vertical edges without enough bottom or top margin is also formalized, including explicit 7×7 examples and their four-sided conjunction. Thus the present margin criterion is a sufficient interior criterion rather than the final every-edge construction. Consequently, the nonexistence of a family of this margin-cover data over all edges of the current open 7×7 graph is also formalized.

- A family of one-edge blocking data determining the uniform every-edge hypotheses: blueprint `thm:peps_normalEdgeBlockingHypotheses_injectiveChain`, issue #2004.
- Edge blocking into red, blue, and complementary injective regions: blueprint `thm:peps_normalEdgeBlockingToInjectiveChain`, issue #1375.
- Application of Lemma `inj_isomorph` after normal blocking: downstream issues #1367, #1368, #1363, and #1364.
- Translationally invariant gauge absorption: blueprint `thm:peps_tiNormalGaugeAbsorption`, issue #1375.
- Final R and S comparison: blueprint `thm:peps_oneVertexComplementComparison`, issue #1360.
- Theorem 3: blueprint `thm:fundamentalTheorem_normalSquarePEPS`, issue #1375.
- General theorem `normal`: blueprint `thm:fundamentalTheorem_normalPEPS`, issue #1375.

5 Blueprint diagrams

The normal route has its own tensor-network diagrams. These are not substitutes for the injective theorem diagrams; they attach the additional normal blocking layer to named proof nodes.[6]

- Lemma `injective_union` shows the four regions $A \setminus B$, $A \cap B$, $B \setminus A$, and $(A \cup B)^c$.
- Definition `peps_normalSquareBlockingRegions` shows the regions R, S, and T from the square-lattice proof.
- Lemma `peps_normalSquareEdgeComplementRegion_topCollar` shows the finite-lattice complementary block A_3 as the local T-region together with the two top-collar rectangles in the normalized 5×7 frame.
- Theorem `peps_normalEdgeBlockingToInjectiveChain` shows the red/blue/complementary blocking around a distinguished edge and the resulting three-site chain.
- Theorem `peps_tiNormalGaugeAbsorption` shows the final translationally invariant local gauge formula with horizontal gauge X and vertical gauge Y.

6 Conclusion

The normal PEPS route is separated from the injective PEPS route in the blueprint and in the GitHub issue tree. The formal proof still remains open: the region-injectivity vocabulary, the four-region decomposition, and the abstract contraction-chain criterion are present, but the tensor-level proof of Lemma `injective_union` is not. The next genuine mathematical steps are that tensor-level union lemma, the source-faithful local and rotated T-injectivity argument, and the every-edge finite-geometry construction. After these normal blocking hypotheses are proved, the downstream proof should reuse the injective edge-insertion route rather than duplicate it. The translationally invariant gauge absorption and the scalar condition remain the final steps of the normal theorem.

References

[MGRPG+18] András Molnár, José Garre-Rubio, David Pérez-García, Norbert Schuch, and J. Ignacio Cirac. Normal projected entangled pair states generating the same state. *New Journal of Physics*, 20(11):113017, 2018.

[6] Chapter 13a calls the public TikZ commands in `blueprint/src/macros/tn_print.tex`. The web blueprint renders cached SVGs from the same commands through `blueprint/src/Packages/tn_diagrams.py`.